

Mercy Naliaka Moraa

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Professional Summary

Software Engineer with strong experience in backend services, full-stack development, and AI-enabled systems. Hands-on expertise in Go, Python, JavaScript, React, REST APIs, Git, Linux, and Docker through intensive, project-driven engineering practice. Skilled in building reliable architectures across local-first applications, Machine Learning workflows, and decentralized tools. Adept at debugging, API integration, cross-layer system design, and translating complex technical requirements into scalable, production-ready software.

Technical Skills

Languages: Go, Python, JavaScript

Frameworks: Flask, FastAPI, Scikit-learn

Tools: Git, GitHub, Linux, Docker, Vercel, Render

Projects

LedgerMate- Financial Tracking system [LedgerMate](#)

Python • Flask • Relational Databases

- Extended LedgerMate with Paystack integration (STK Push + C2B webhooks), automatically posting confirmed mobile-money payments to the ledger and eliminating manual reconciliation of M-Pesa transactions.
- Used 14 REST API endpoints across inventory, payments, and financial reporting, with idempotent webhook handling and functional test scripts covering auth, inventory flows, and statement generation.
- Designed an inventory module with weighted-average cost tracking, integrating stock purchases and sales into the ledger, and built automated Profit & Loss and Balance Sheet generation from transactional data ;including fixing a cost double-counting bug to ensure Assets = Liabilities + Equity.

DigiCow AI - Farmer Intelligence System [DigiCow AI](#)

Python • FastAPI • Neo4j • GraphRAG

- Engineered a Neo4j knowledge graph database modeling complex relationships between smallholder profiles, soil parameters, livestock history, and localized diseases to power multi-variable context queries for decision support.
- Implemented a GraphRAG retrieval pipeline linking Neo4j graph context with Featherless AI, grounding LLM outputs in official KALRO and ILRI research manuals to eliminate hallucinations and generate bilingual advisories in English and Swahili.
- Built and optimized FastAPI backend endpoints to serve low-latency field advisories, priority-scored agent task queues, and structured follow-up logs for extension agents.

Work Experience

QA Tester (Contract) — JaGedo Aug 2026

- Performed manual QA testing across web platforms for a digital marketplace application, covering multiple user roles and workflows.
- Logged, prioritized, and tracked defects using a project management tool, documenting clear reproduction steps, expected vs. actual behavior, and supporting evidence.
- Validated features against documented business requirements and collaborated with developers and product managers to clarify requirements and confirm intended behavior.

Software Engineer — Zone01 Kisumu Jun 2026 – Present

- Build backend applications in Go, applying clean code principles and modular design.
- Solve algorithmic problems and manage version control workflows using Git and GitHub.
- Collaborate with peers on production-style projects, following software engineering best practices.

Industrial Attachment — Maseno University

- Provided IT support and maintained systems and networks for the university.

Education

Bachelor of Science in Information and Communication Technology Management - Maseno University